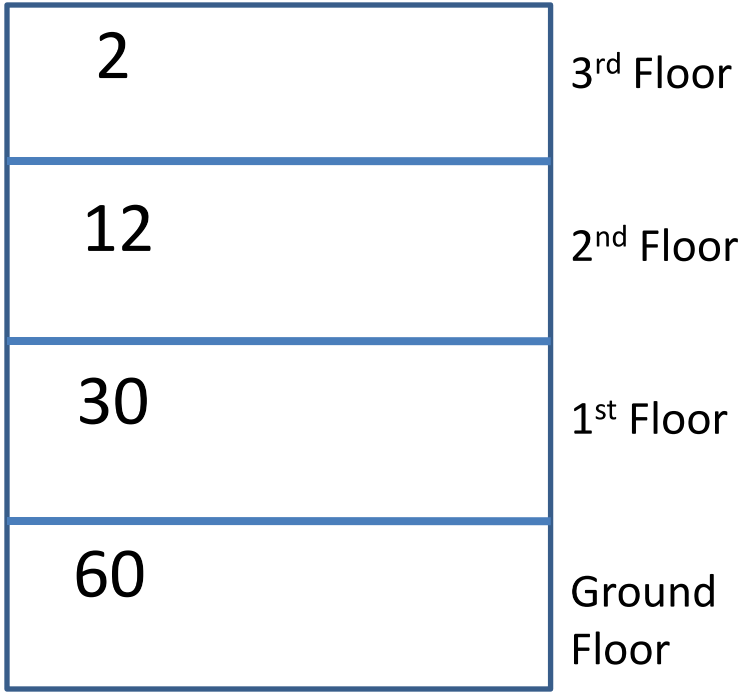
VLSM – Variable Length Subnet Mask

Email: **Need 2, 12, 30 and 60 IP Addresses for Departments**.

Ans: In case of VLSM,
We consider and will
allot IP Addresses on
the basis of IP Addresses
required For Separate
departments like 60 in
Ground the 30 in 1st
12 in 2^{nd} and 2 in 3^{rd} Floor.



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For Ground Floor:

Step 1: Find the Host bit "n":

(2^n)-2>=60 Satisfy Equation by putting

(2^6)-2>=60 values of N as 0 1 2 3 4 5 6 7 8

62>=60, here n = 6

Last 6 Bits of 32 Bits will be used as host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

Default Subnet Mask: **255.255.255.0**

11111111.11111111.11111111.00000000

Moving Right to Left and turn 6 bits "0" and Remaining 1

11111111.11111111.11111111.11000000

Customized Subnet Mask: **255.255.255.192**

CIDR Value = /26

2	3rd Floor
12	2nd Floor
30	1st Floor
60	Ground Floor

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Step 3: Block Size = 2^n = 2^6 = 64

Allotting the IP Address:

Ground Floor: Required = 60 IP Addresses,

Wastage = 0

192.168.10.0/26 **- Network IP**

192.168.10.1/26

192.168.10.2/26

192.168.10.3/26

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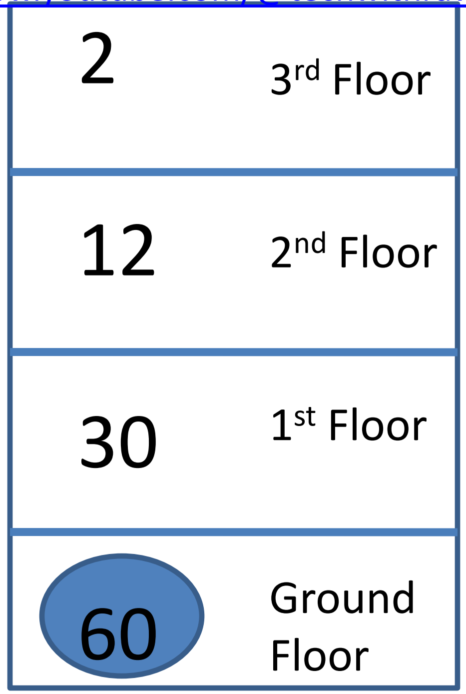
Valid IP Addresses = 62

192.168.10.62/26

192.168.10.63/26 **- Broadcast IP**



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For 1^{st} Floor:

Step 1: Find the Host bit "n":

(2^n)-2>=30 Satisfy Equation by putting

(2^6)-2>=30 values of N as 0 1 2 3 4 5 6 7 8

30>=30, here n = 5

Last 5 Bits of 32 Bits will be used as host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

Default Subnet Mask: **255.255.255.0**

11111111.11111111.11111111.00000000

Moving Right to Left and turn 5 bits "0" and Remaining 1

11111111.11111111.11111111.11100000

Customized Subnet Mask: **255.255.255.224**

CIDR Value = /27

2	3^{rd} Floor
12	2^{nd} Floor
30	1^{st} Floor
60	Ground Floor

Step 3: Block Size = 2^n = 2^5 = 32

Allotting the IP Address:

1st Floor Need 30 IP Addresses, Wastage is 0

192.168.10.64/27 **- Network IP**
192.168.10.65/27
192.168.10.66/27
192.168.10.67/27

Valid IP Addresses = 30

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192.168.10.94/27
192.168.10.95/27 **- Broadcast IP**

2	3rd Floor
12	2nd Floor
30	1st Floor
60	Ground Floor

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For 2nd Floor:

Step 1: Find the Host bit "n":

(2^n)-2>=12 Satisfy Equation by putting

(2^4)-2>=12 values of N as 0 1 2 3 4 5 6 7 8

14>=12, here n = 4

Last 4 Bits of 32 Bits will be used as host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

Default Subnet Mask: **255.255.255.0**

11111111.11111111.11111111.00000000

Moving Right to Left and turn 4 bits as "0" and other 1

11111111.11111111.11111111.11110000

Customized Subnet Mask: **255.255.255.240**

CIDR Value = /28

2	3rd Floor
12	2nd Floor
30	1st Floor
60	Ground Floor

Step 3: Block Size = 2^n = 2^4 = 16

Allotting the IP Address:

2nd Floor Need 12 IP Addresses, Wastage = 0

192.168.10.96/28 **- Network IP**

192.168.10.97/28

192.168.10.98/28

192.168.10.99/28

Valid IP Addresses = 14

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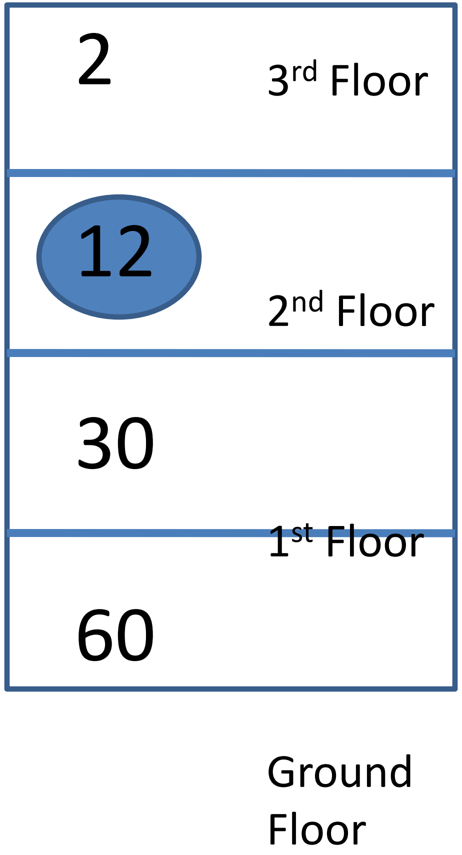
192.168.10.110/28

192.168.10.111/28 **- Broadcast IP**



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For 3rd Floor:

Step 1: Find the Host bit “n”:

(2^n)-2>=2 Satisfy Equation by putting

(2^2)-2>=2 values of N as 0 1 2 3 4 5 6 7 8

2>=2, here n = 2

Last 2 Bits of 32 Bits will be used as host Bit.

STEP 2: Default Subnet Mask to Custom Subnet mask

Default Subnet Mask: **255.255.255.0**

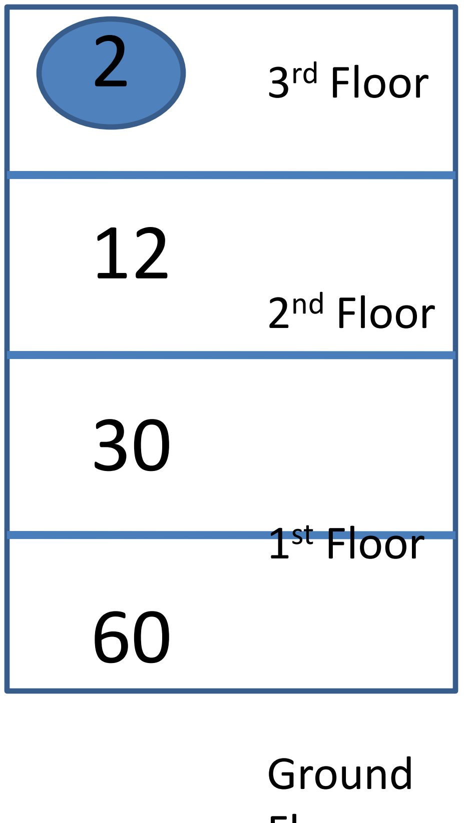
11111111.11111111.11111111.00000000

Moving Right to Left and turn 2 bits “0” and Remaining 1

11111111.11111111.11111111.11111100

Customized Subnet Mask: **255.255.255.252**

CIDR Value = /30



Step 3: Block Size = 2^n = 2^2 = 4

Allotting the IP Address:

3rd Floor Need 2 IP Addresses, Wastage = 0

192.168.10.112/30 **- Network IP**

192.168.10.113/30 **Valid IP Addresses = 2**

192.168.10.114/30

192.168.10.115/30 **- Broadcast IP**

2	3rd Floor
12	2nd Floor
30	1st Floor
60	Ground Floor

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Total Wastage:

Now we have no wastage of IP Addresses in each department:

Ground Floor: 62 – 60 = 2 IP

1^{st} Floor: 30 - 30 = 0 IP

2^{nd} Floor: 14 - 12 = 2 IP

3^{rd} Floor: 2 – 2 = 0 IP

Total Wastage

2+0+2+0 = 4

(Cannot be saved)

Required IP = 2, (2 Allotted),	3^{rd} Floor
Required IP = 12, (14 Allotted)	2^{nd} Floor
Required IP = 30, (30 Allotted)	1^{st} Floor
Required IP = 60, (62 Allotted)	Ground Floor